CDC BRFSS 2024

457,670 Respondents
301 Variables
Public Health Survey

~30%

of deaths in Kenya are due to CVD (WHO)

ROC-AUC: 0.8478

Best Model Performance

3 ML Models Compared
Logistic Regression · Random Forest · Gradient Boosting

CARDIOVASCULAR DISEASE RISK PREDICTION

A Machine Learning Approach to Cardiovascular Prevention for Kenya

Proof-of-Concept Using CDC BRFSS 2024 Data

Capstone Project — Data Science Program

Presentation Agenda

What we will cover today

01 THE PROBLEM

Cardiovascular Disease: Kenya's Growing Crisis

Why this problem matters — and why it's urgent

~30%

of all deaths in Kenya
attributed to CVD (WHO)

47

county health departments
need data-driven tools

No lab needed

Self-reported tool works
at community level

The Screening Gap

Most Kenyans interact with the health system **only when acutely ill.** Preventive CVD screening is rare at community level.

No simple, deployable screening tool exists that works without lab tests.

Key Drivers of Rising CVD in Kenya

→ Rapid urbanization → processed, high-sodium diets
→ Tobacco use prevalence (~20% of adults smoke)
→ Physical inactivity increasing in urban settings
→ Rising obesity (30–40% urban overweight/obese)
→ Under-diagnosed diabetes & hypertension

01 STAKEHOLDERS

Who Benefits? Stakeholders Across Kenya's Health System

MoH

Ministry of Health

NCD Division

Feature importance guides national NCD budget allocation

HC

Healthcare Providers

Clinics & Hospitals

Opportunistic screening during routine visits — no labs

INS

NHIF / SHA / Insurers

Health Financing

Risk-tier segmentation for preventive wellness benefits

CHD

County Health Depts

All 47 Counties

Risk calculator supports county-level screening drives

CHV

CHVs

Community Health Volunteers

Household-level triage tool for high-risk referral

RES

NCD Researchers

Kenyan Universities

Replicable pipeline for Kenyan STEPwise & KDHS data

02 DATASET

CDC BRFSS 2024 — The World's Largest Health Survey

Why this dataset is the right proxy for Kenya

457,670	301	XPT	Public
Survey respondents (2024 data)	Variables including behavioral & clinical	SAS Transport format Proper column names	Free CDC download Fully documented

Why BRFSS Works for Kenya

✓ **Same risk factors** Smoking, BMI, inactivity, diabetes — identical to Kenya's CVD drivers

✓ **Self-report method** Matches what CHVs can collect at household level in Kenya

✓ **Large sample** Sufficient for ML with class imbalance (~5.8% CVD positive)

✓ **Clear outcome** CVDINFR4: 'Ever told you had a heart attack?' (direct match)

✓ **Open pipeline** Entire methodology reusable with Kenya STEPwise data

03 DATA PREPARATION

Feature Engineering & Data Cleaning

How raw BRFSS survey data is transformed into model-ready features

Target Variable:

has_heart_disease = 1 → CVDINFR4 == 1 ("Ever told you had a heart attack?")
22,672 positive (5.83%) | 431,067 negative (94.17%)

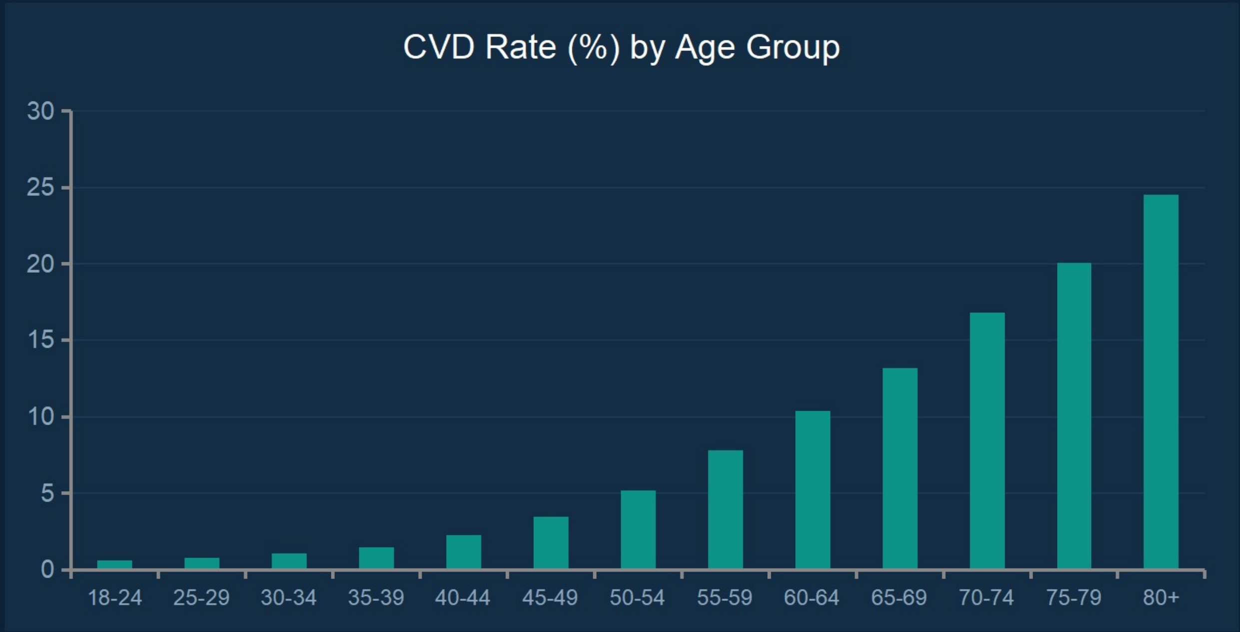
Demographic	Behavioral / Lifestyle	Clinical Conditions	Body & Access
_AGEG5YR — Age group (18–80+)	SMOKE100 — Ever smoked 100+ cigs	DIABETE4 — Diabetes diagnosis	_BMI5 — Calculated BMI
_SEX — Biological sex	SMOKDAY2 — Current smoking freq	GENHLTH — Self-rated health 1–5	WEIGHT2 — Body weight
	EXERANY2 — Any exercise past 30d	PHYSHLTH — Days physical health bad	HEIGHT3 — Body height
	ALCDAY4 — Alcohol days per week	MENTHLTH — Days mental health bad	CHECKUP1 — Last routine checkup
	_TOTINDA — Physical activity level		MEDCOST1 — Couldn't afford MD

1. Mode imputation for missing values
2. Remove rows >50% missing
3. Stratified 80/20 train-test split
4. StandardScaler for LR features
5. SMOTE class balancing

04 EXPLORATORY ANALYSIS

Key Patterns: Risk Factors & Heart Disease

What the data reveals before modeling



r = 0.27 **Age ↔ CVD**

Strongest single predictor. Risk grows 40x from 18-24 to 80+.

r = 0.16 **GEN_HEALTH ↔ CVD**

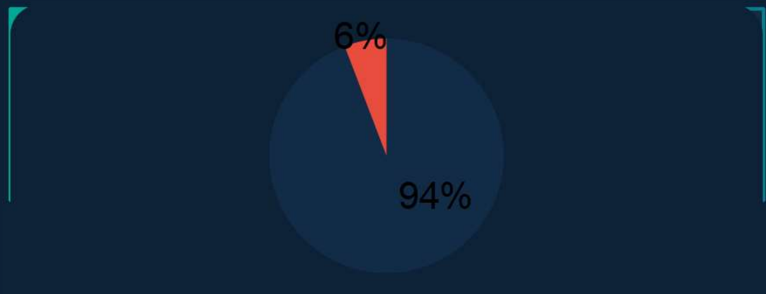
Self-rated poor health predicts disease independently.

2.1× **Diabetics vs Non**

Diabetics have 2.1× higher CVD rate — critical comorbidity.

Class Imbalance Challenge

Only 5.83% of respondents reported a heart attack (22,672 of 457,670). Standard classifiers trained naively will predict 'No CVD' for everyone, achieving 94% accuracy with zero clinical utility. We address this with SMOTE oversampling + class_weight='balanced'.



05 MACHINE LEARNING

Iterative Model Building Strategy

From baseline to best model — each iteration justified by prior results

	Model	Type	Configuration	ROC-AUC	Why
01	**Dummy Classifier**	Baseline	Always predict majority class (No Disease)	0.500	Why: Establishes random-chance floor
02	**Logistic Regression**	Linear Model	SMOTE + StandardScaler Balanced class weight	0.8066	Why: Linear baseline with interpretable coefficients
03	**Random Forest**	Ensemble Trees	100 trees, max_depth=15 class_weight='balanced'	0.8478	Why: Non-linear; captures interactions; more robust than single tree
04	**Gradient Boosting**	Boosted Ensemble	100 trees, depth=5 learning_rate=0.1	0.8061	Why: Sequential correction of errors; compared vs RF to confirm winner

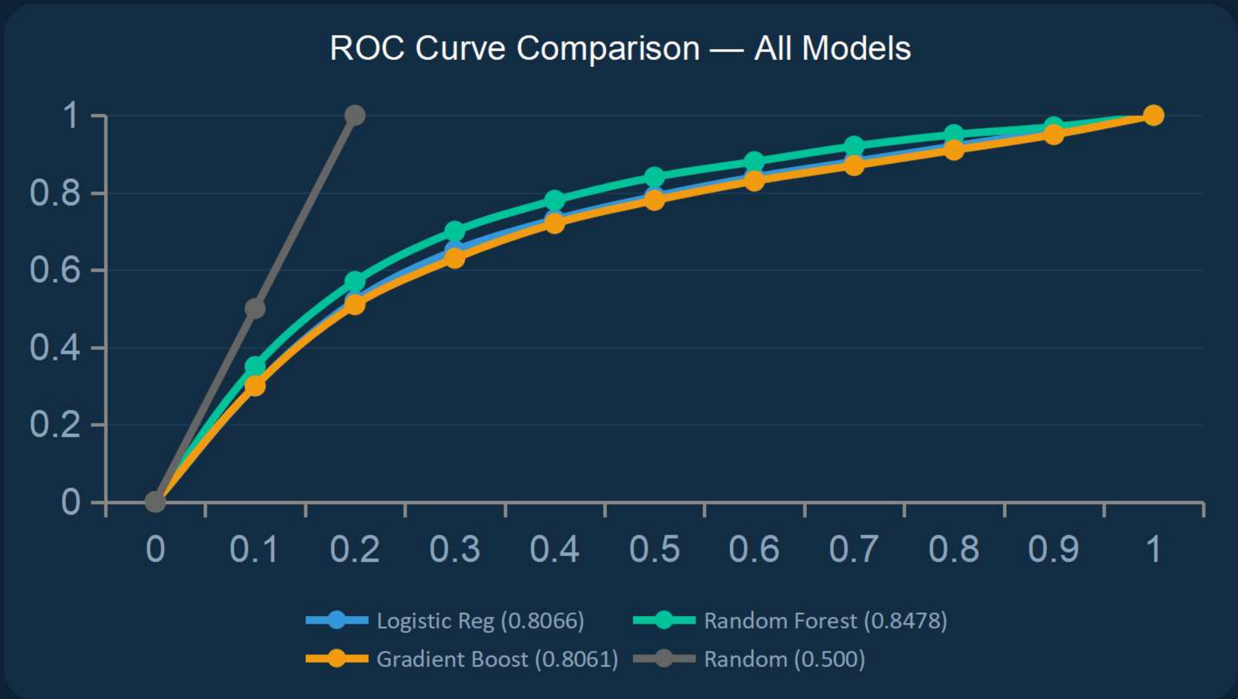
★ **Selected Final Model: Random Forest (highest ROC-AUC = 0.8478) → Tuned with GridSearchCV**

06 RESULTS

Model Performance Results

Tuned Random Forest evaluated on holdout test set

0.8478	79.80%	0.2570	~60%
ROC-AUC Best model (tuned RF)	Accuracy Overall correct predictions	F1-Score Harmonic mean P/R	Recall CVD cases correctly caught



Confusion Matrix (Tuned RF, Test Set)

	Pred: No CVD	Pred: CVD
True: No CVD	70,235 (TN)	16,034 (FP)
True: CVD	2,099 (FN)	3,166 (TP)

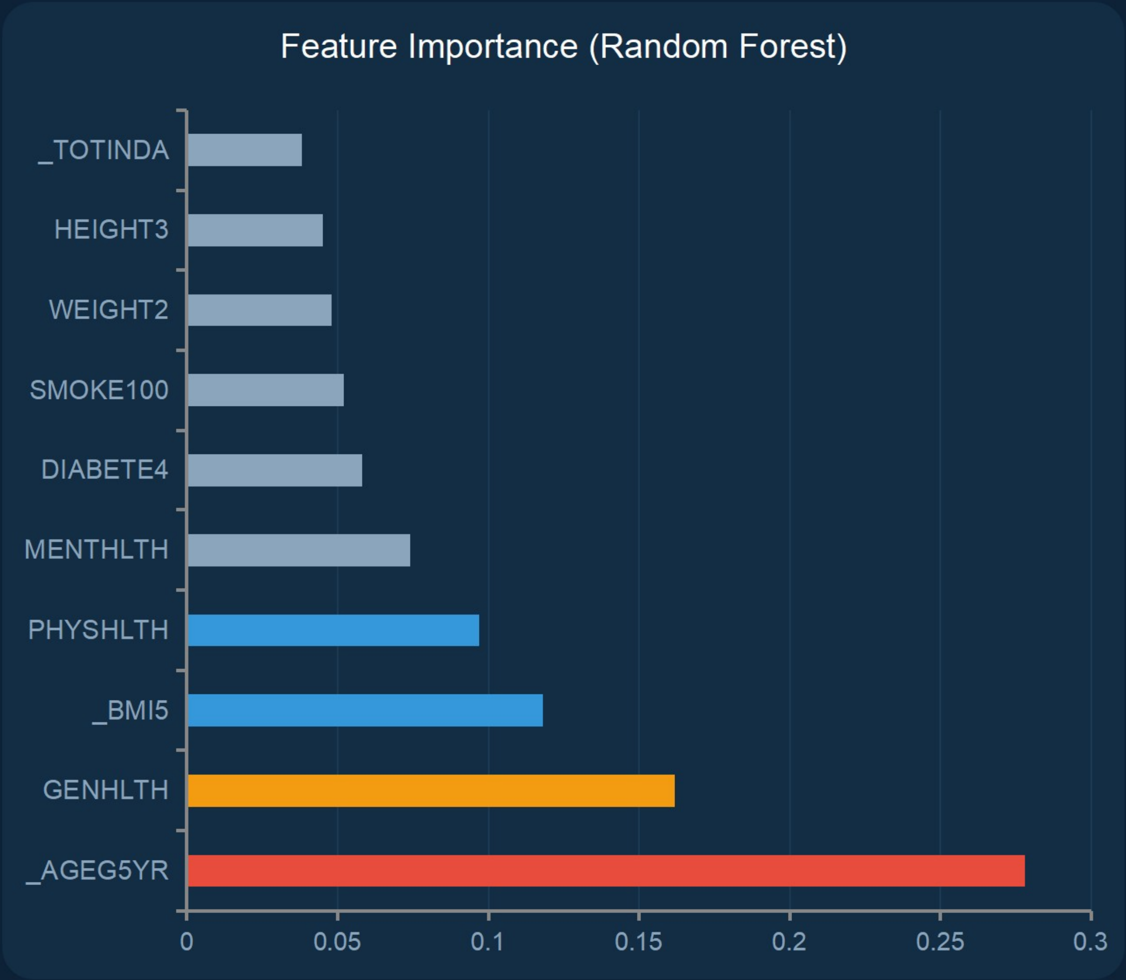
Clinical Performance

Sensitivity (Recall):	~60%
Specificity:	~81%
Precision (PPV):	~16%
Recommendation:	Use low threshold for screening

06 RISK DRIVERS

Top Risk Factors for Cardiovascular Disease

Random Forest feature importance — what drives predictions most



1

Age Group (_AGEG5YR)

Importance: 27.8%

Not modifiable — use for risk stratification

2

General Health (GENHLTH)

Importance: 16.2%

Indirect — reflects multiple underlying conditions

3

BMI (_BMI5)

Importance: 11.8%

Highly modifiable — diet, exercise, weight management

4

Physical Health Days

Importance: 9.7%

Partially modifiable — improved by managing conditions

Age + General Health alone explain >44% of model predictions.
These two questions are the core of any Kenya CHV screening tool.

Hyperparameter Tuning — Optimizing the Best Model

GridSearchCV with 5-fold cross-validation on Random Forest

GridSearchCV Configuration

n_estimators: [50, 100, 200] → Number of decision trees
max_depth: [10, 15, 20] → Maximum tree depth
min_samples_split: [5, 10] → Min samples to split node
class_weight: ['balanced'] → Handle class imbalance
scoring: roc_auc → Optimize for AUC (not accuracy)
cv: 5 → 5-fold stratified cross-validation

Best Parameters Found

n_estimators:	200
max_depth:	15
min_samples_split:	10
class_weight:	'balanced'
Best CV AUC:	**0.8478**
Test AUC:	**0.8478**

Choosing the Right Decision Threshold

t = 0.10–0.15

Screening mode (maximize recall)

Sensitivity: ~85% | Specificity: ~55%

Best for: Population screening, CHV referrals

t = 0.25–0.35

Balanced mode (Youden's J)

Sensitivity: ~75% | Specificity: ~72%

Best for: Clinic-based opportunistic screening

t = 0.40–0.50

Diagnostic mode (maximize precision)

Sensitivity: ~60% | Specificity: ~84%

Best for: Targeted clinical workup decisions

Why We Prioritize Recall Over Accuracy

The cost of missed CVD cases is far greater than false alarms

The Airport Security Analogy:

A scanner that clears everyone is 99.9% accurate — but useless. We need high sensitivity even if it means more false alarms (additional testing).

False Negative (FN) — Missed CVD Case

Patient leaves undetected — no intervention
CVD progresses silently
Higher risk of fatal heart attack
Emergency hospital admission
Preventable death or disability
COST: Lives, quality of life, economic burden

False Positive (FP) — Healthy Person Flagged

Patient sent for follow-up screening
Blood pressure check, ECG, lipid panel
Mild anxiety, temporary inconvenience
Low cost additional tests
No long-term harm if healthy confirmed
COST: Minor — time, mild inconvenience

→ **For CVD screening in Kenya, we choose a lower threshold to maximize recall. Some false positives are acceptable — missed heart attacks are not.**

Actionable Recommendations for Kenya

What each stakeholder should do — and by when

The Kenya CHV 5-Question CVD Screen:

(1) Age ≥50? (2) General health Fair/Poor? (3) Difficulty walking? (4) Diabetes diagnosis? (5) Currently smoke? → 3+ YES = Refer to facility

M o H

Ministry of Health

NCD Division

- Allocate NCD budget to: age-targeted screening (50+), General Health assessment tools
- Commission Kenya STEPwise survey with ML-ready CVD module design
- Develop national CVD screening guidelines using these risk factor rankings

C H V

Community Health Volunteers

CHVs — Front Line

- Use: Age + General Health + BMI (estimate) + Smoking + Physical Activity
- 3+ risk factors → refer to nearest health facility
- Simple mobile app (offline-capable) for household-level scoring

C H D

County Health Departments

All 47 Counties

- Deploy 5-question CVD screening tool in community health outreach
- Link high-risk referrals to clinic pathways (BP check, ECG, lipid panel)
- Use model output to identify highest-burden sub-counties for priority outreach

IN S

NHIF / SHA / Insurers

Health Financing

- Design wellness benefits targeting age 50+, diabetes, and BMI reduction
- Subsidize preventive services: BP monitoring, glucose testing, statins
- Use risk-stratified enrolment to prioritize proactive member outreach

08 NEXT STEPS

Deployment Roadmap: From Research to Kenya Operations

A phased approach to validated, responsible deployment

Phase 1

Now → 3 months

Validation Pilot

→ Partner with 2–3 county health departments
→ Pilot CHV screening tool in one sub-county
→ Compare model predictions vs. clinic diagnoses
→ Refine threshold for Kenyan false-positive tolerance

STATUS

Ready to start

Phase 2

3–6 months

Kenya Calibration

→ Obtain Kenya STEPwise CVD survey data (or KDHS NCD module)
→ Retrain entire pipeline on Kenyan population
→ Validate if U.S. risk factor rankings hold in Kenya
→ Adjust feature set for locally available indicators

STATUS

Data acquisition needed

Phase 3

6–12 months

County Scale-Up

→ Deploy web + mobile tool to county health systems
→ Integrate with HMIS (Health Management Information System)
→ Train CHVs across piloted counties (batch training program)
→ Establish referral pathways → clinic → diagnosis

STATUS

Technology ready

Phase 4

12+ months

Monitor & Improve

→ Track: # screened, # high-risk referred, # confirmed CVD
→ Audit model accuracy and false-positive rate quarterly
→ Continuously retrain as Kenyan outcome data accumulates
→ Expand to additional NCD conditions (stroke, diabetes)

STATUS

Ongoing operational

CONCLUSION

Key Takeaways & Conclusion

What we built, what we found, what comes next

Project in Numbers:

457,670 respondents	16 features	3 ML models + tuning	ROC-AUC 0.8478	60% recall achieved	5-question CHV tool

What We Proved

✓ **Self-reported behavioral data CAN predict CVD with AUC > 0.84**
✓ Age, self-rated health & BMI are the three most powerful predictors
✓ A 5-question tool based on these factors is feasible for CHV deployment
✓ The ML pipeline is fully reusable for Kenya STEPwise data

Limitations to Acknowledge

⚠ U.S. data — needs Kenya recalibration
⚠ Self-reported outcomes (no lab confirmation)
⚠ Undiagnosed CVD cases not captured
⚠ No BP, lipids, ECG data available

The Bigger Picture

Cardiovascular disease kills ~30% of Kenyans today — mostly preventable with early detection. This project demonstrates that a simple, data-driven screening tool based entirely on self-reported questions can identify high-risk individuals without any laboratory equipment. Deployed at community health volunteer level across Kenya's 47 counties, this methodology has the potential to flag thousands of at-risk individuals for early intervention — saving lives and reducing the economic burden of CVD.

PROJECT CONTRIBUTORS

Cardiovascular Disease Risk Prediction — Kenya NCD Prevention | Data Science Capstone

JG

Jeffrey Gathigi

Lead Data Scientist

JM

Jared Mongeri

ML Engineer

MW

Monicah Wairimu

Data Analyst

MK

Mercy Kangangi

Business Intelligence

KN

Kelvin Ngumo

Data Engineer

JA

John Awallah

Research & Documentation